TANZANIA HEADS OF ISLAMIC SCHOOLS COUNCIL

FORM IV INTER ISLAMIC MOCK EXAMINATION 2021

032/2A **CHEMISTRY 2A**

ACTUAL PRACTICAL

MARKING SCHEME

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1. Volume of acid:

	Pilot	T_1	T_2	T_3
Final volume (cm^3)	16.50	32.60	48.50	36.00
Initial volume (cm^3)	0.00	16.50	32.60	20.00
Volume used (cm^3)	16.50	16.10	15.90	16.00

Table 06 marks @ correct titre value (01) mark two decimal places (01) mark

(a) The average volume of Q = $\frac{(T_1+T_2+T_3)cm^3}{3}$

$= \frac{(16.10+15.90+16.00)cm^3}{3}$ **01 mark**

$\underline{= 16.00cm^3.}$ **01 mark**

(b) 20 cm^3 of solution P required 10.8cm^3 of solution Q for complete reaction. **01mark**

(c) Balanced chemical equation for the reaction. **02marks**

$M_2CO_3(aq) + 2HCl_{(aq)} \rightarrow MCl_{(aq)} + H_2O_{(l)}$ **02marks**

Data given:

$CO^{2-}_{3(aq)} + 2H^{+}_{(aq)} \rightarrow CO_2(g) + H_2O_{(l)}$ **02marks**

(d) Molarity of Acid (Ma) = 0.25M

Volume of Acid used (Va) = 16.00cm^3.

Molarity of Base (mb)= ?

Number of moles of acid (na) = 2

Number of moles of base (nb) = 1

Mass of base 4.24g in 0.5 litres. **00 ½ mark**

From the formula $\frac{MaVa}{Mbvb} = \frac{na}{nb}$ **00 ½**

$Mb = \frac{MaVa\ nb}{Vbna}$ **00 ½**

$Mb = \frac{0.25\times16.x1}{25\times2}$ **01mark**

Molarity of M_2CO_3 (Mb) = 0.08M. **01Marks**

Molar mass of M_2CO_3:

Molar mass = $\frac{Conc.in\ g/<\ of\ m_{2CO_3}}{Molarity}$ **01**

But conc. In g/< of m_2CO_3;

= $\frac{Mass}{Volume\ of\ a\ solution}$ **01**

= $\frac{4.24g}{0.5dm^3}$ **01**

= 8.48g/< **01**

Molar mass = $\frac{8.48g/<}{0.08M}$ **01**

∴ Molar mass = 106/mol of m_2CO_3. **01**

Molar mass of m (R.A.M) of m.

M_2CO_3 = 106

2M+12+3(16) = 106

2M+ 60 = 106 **01**

2M = 46

M = $\frac{46}{2}$=23 **00 ½**

∴ Element M is sodium. **01**

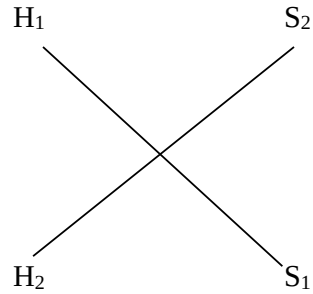
2. (i)

Temperature ^{o}C	**Time (s)**	**Rate S^{-1}**
50	30	0.03
60	24	0.04
70	18	0.06
80	14	0.07

Table = 05 marks @ = 00 ½ for column of time and rate

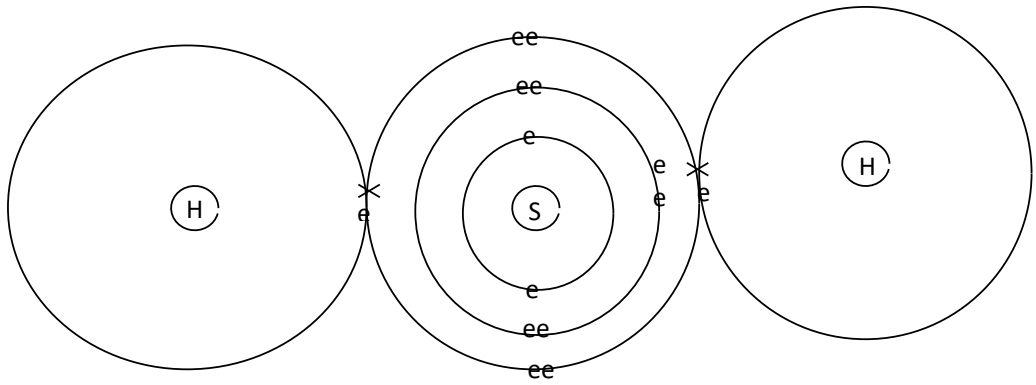
(ii) In order to avoid splitting of hot solution i.e. flying out of the piece of hot solutions. **01mark**

(iii) The product which carries the solution to be cloudy is sulphur. Sulphur joined with hydrogen through sharing of electrons to form hydrogen sulphide. **01mark**



(01mark)

H_2S – Hydrogen sulphide



03

Diagram showing chemical bonding (H_2S)

Whereby X-electron of H

e – electrons of S

2. (iv) From

$Na_2S_2O_3(aq) + 2HCl(aq) \rightarrow 2NaCl(aq) + S(s) + H_2O(l) + SO_2(g)$ **02**

01mark

$2H^+_{(aq)} + S_2O^{2}_{3(aq)} \rightarrow S_{(s)} + H_2O_{(l)} + SO_{2_{(g)}}$ **02marks**

(v) Consider the graph

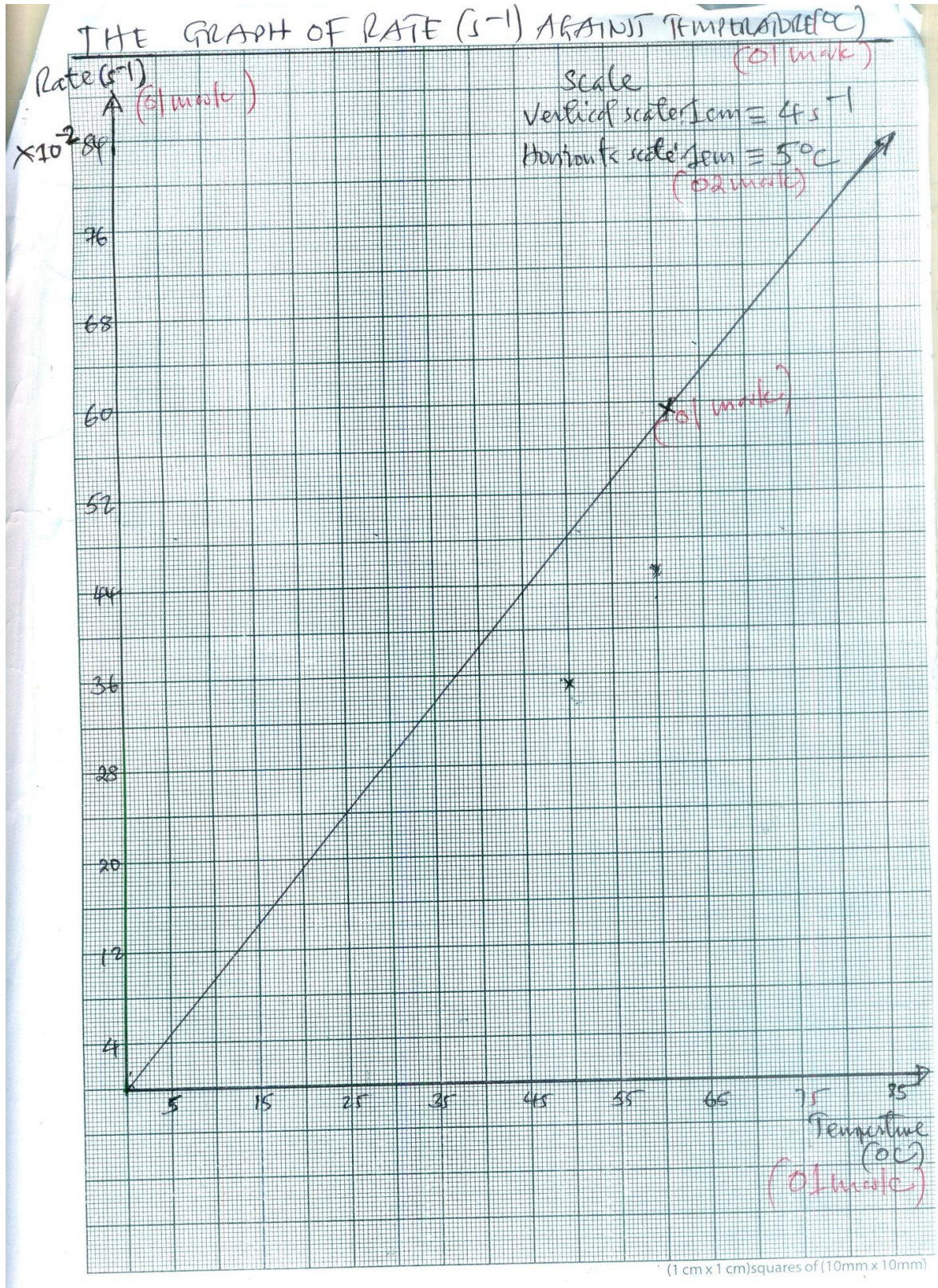
(vi) The nature of the graph is straight line, this signify that the rate of reaction increases as an increase in temperature in endothermic reaction. **03marks**

NB:

(v)
- Heading - **01mark**
- Scales - **02marks**
- y-plane - **01mark**
- x-plane - **01mark**

Point located/nearby straightline - **@ = 01**

Total 04 marks


THE GRAPH OF RATE (s^{-1}) AGAINST TEMPERATURE (°C)
(01 mark)
Rate (s^{-1})
(01 mark)
$\times 10^{-2}$
Scale
Vertical scale: 1cm = 4 s^{-1}
Horizontal scale: 1cm = 5°C
(02 mark)
84
76
68
60
52
44
36
28
20
12
4
(01 mark)
5
15
25
35
45
55
65
75
85
Temperature (°C)
(01 mark)
(1 cm x 1 cm)squares of (10mm x 10mm)